

FAR WESTERN UNIVERSITY
Semester End Examination-2081
Parallel Computing (CSIT.421)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. Explain divide and conquer technique.
2. Differentiate between data and task parallelism.
3. Define parallel programming.
4. Explain Conjugate gradient method.
5. Define Karp-Flatt Metric.
6. What is OpenMP?
7. Explain about Loop Unrolling.
8. Define hyper quicksort.

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. Explain the Amdahl's law for measuring speed up performance with the help of an example.
10. Explain Monte Carlo methods. Write down its application.
11. What is shared memory model? Write its advantages and disadvantages.
12. What is message passing model? Write down its advantages and disadvantages.
13. Use Eratosthenes algorithm to find all the prime number between 1 to 30.
14. Write short notes on: (any two)
 - a. Isoefficiency Metric
 - b. Gustafson Barsis Law
 - c. Fast Fourier Transform (FFT) algorithms

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. What do you mean by combinatorial search? Explain any one example in detail.
16. Define Foster's design methodology. Explain its steps in detail.
17. What is pipeline and how it works? Explain about Flynn's taxonomy.
18. Write Floyd-Warshall algorithm. Trace it with suitable example.

FAR WESTERN UNIVERSITY
Semester End Examination-2081
Advanced Database Design (CSIT.423.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Full Marks: 100

Semester: Eighth

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. How do you measure performance of disks?
2. What are the functions of Buffer manager?
3. How does B-tree differs from a B+ tree?
4. What are the factors for evaluating indices?
5. What are two differences between DBMS and RDBMS?
6. Define Primary and candidate key with examples.
7. Define hash function.
8. What is normalization?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. What are the concurrency control problems that may occur in database?
Describe two phase locking with example.
10. How database failure occurs? How do you recover database?
11. How does Bucket overflow occurs and How do you handle it?
12. What is indexing? Explain primary indexing with diagram and example.
13. How does DBMS overcome the problems of file processing system?
14. Write short notes on (any two)
 - a. Query Optimization
 - b. Fixed-length record and variable-length record.
 - c. Shadow paging

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Discuss the properties of transaction in DBMS. Explain the states through which a transaction passes during execution. (4+8)
16. Explain RAID technology up to (0 and 1) level in database. Also mention its importance.
17. Why is lossless-join important? Explain fifth normal form (5 NF) with examples.
18. What is Query processing? What are the steps involved in query processing? Explain with examples.

FAR WESTERN UNIVERSITY
Semester End Examination-2081
E-Governance (CSIT.525.2)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Full Marks: 100

Semester: Eighth

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. What are the key objectives of e-governance?
2. Describe the concept of digital divide in e-governance.
3. What is Open Government Data (OGD)?
4. How can e-governance improve public service delivery?
5. Explain government to the business model.
6. How e-governance security difficulties can be withdrawn?
7. What is e-readiness in e-governance?
8. Define data system infrastructure.

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. Explain in brief about scope and content of e-governance.
10. Differentiate between e-governance and e-government. Explain the issues in e-governance implementations and how to solve these issues?
11. Explain the components of data warehouse. What are applications of data mining in e-governance in context of Nepal?
12. What are pre-requisites of e-readiness for intra structure development?
13. How can we achieve good governance through e-governance model? Explain.
14. Short note on
 - a. IT in judiciary
 - b. Electronic service delivery

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Why do we need IT infrastructure to implement E-Governance? Explain network and computing infrastructure needed to implement E-governance.
16. What is E-Governance model? Explain broadcasting/wider dissemination model and critical flow model in detail.
17. Why do we need IT infrastructure to implement E-Governance? Explain network and computing infrastructure needed to implement E-governance.
18. Explain human infrastructure preparedness and technical infrastructure preparedness for e-governance.

FAR WESTERN UNIVERSITY
Semester End Examination-2081
Cloud Computing (CSIT.424.4)

Faculty: Science and Technology (CSIT)

Level: Undergraduate

Semester: Eighth

Full Marks: 100

Time: 3hrs.

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Group – 'A'

Attempt all questions (Very short answer questions)

8×3 = 24

1. What are the main features of Cloud Computing?
2. What is *Elastic Compute Cloud (EC2)*?
3. How does the monitoring agent monitor cloud usage?
4. How does cloud-native foundation define Cloud-native applications?
5. What are major differences between traditional corporate data center and cloud data center?
6. How resource replication takes place in Cloud Computing?
7. What is the platform as a service?
8. What is the difference between scalability and elasticity?

Group – 'B'

Attempt any five questions. (Short answer questions)

5×8=40

9. In what manner does parallel and distributed computing differ?
10. What are the implementation issues on SAAS? Explain.
11. Explain the data center virtualization.
12. Explain the disaster recovery planning of cloud computing.
13. Describe possible services that can be achieved through Infrastructure-as-a-Service (IaaS).
14. What do you mean by Elastic IP Addressing? Describe how Elastic IPs work in cloud services.

Group – 'C'

Attempt any three questions. (Long answer questions)

3×12 = 36

15. Explain the need of virtualization. Explain external network virtualization in different layers.
16. Describe features, challenges and benefits of cloud computing.
17. Explain the building blocks required to build a data center.
18. What is Service Oriented Architecture (SOA)? Explain the principal characteristics of SOA.